

CSCI 5502 - Data Mining Group Project Poster Presentation

April 29th, 2025

Instructions

- One slide per Group
 - Include Group #, Member's Name, URL (Webpage and GitHub)
 - One or more images (webpage screenshot or any other relevant image)
 - Feel free to modify the style for your slide
- Limit to 8 minutes of presentation. Discuss
 - Project Overview and motivation (Milestone 1)
 - Challenges in Data Collection/Processing (Milestone 2)
 - Challenges in Data Modeling (Milestone 3)
 - What did you learn from this project?
- The presentation order will be random.
- On April 29th, you will perform a peer review evaluation as part of the last individual in-class assignment (Canvas). This presentation will help you promote your project.

INTRODUCTION

Since 1999, over 760,000 people have died from drug overdoses, with opioids like fentanyl driving a sharp rise in deaths. This research explores trends in opioid misuse, demographic and geographic factors, and the influence of pharmaceutical companies. Despite efforts like naloxone distribution and reduced prescribing, over 220 people die each day, and treatment gaps remain. The crisis impacts healthcare, law enforcement, social services, and vulnerable groups. Using data mining and machine learning, this project aims to uncover patterns and guide solutions to reduce deaths and support communities.

OBJECTIVES

Answer the following questions:

- What states have the highest rates of opioid prescriptions?
- How are prescription rates changing over time?
- What states have the highest rates of opioid overdoses?
- What drug classes are experiencing higher overdose rates?
- Is opioid use correlated to patient demographics such as income, mental health status, and education?
- How are overdose rates and access to healthcare correlated?

METHODS & RESULTS

Dataset 1: Opioid Prescription Rates and Patterns by State and Provider Type

A decision tree predicted with 76% accuracy the credentials of a provider (for example MD, PA, or DDS) based on their prescribing habits.

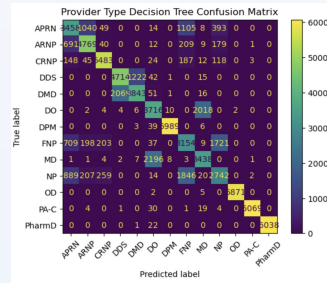
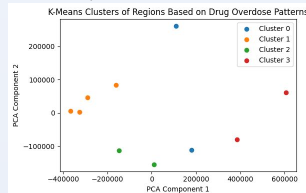


Figure 1. Decision Tree Confusion Matrix

Dataset 2: Drug Overdose Rates by Location and Drug Type in 2019

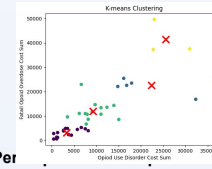
K-means clustering of dataset 2 revealed regional overdose patterns: high fentanyl and heroin deaths in the Northeast and Midwest, methamphetamine in the West and South, low counts in rural states, and diverse drug overdoses in urban areas. Nearby regions often clustered together. Apriori analysis showed frequent co-occurrence of fentanyl with heroin (Northeast/Appalachia), methamphetamine (West/South), and Alprazolam (Southeast/Midwest).



METHODS & RESULTS

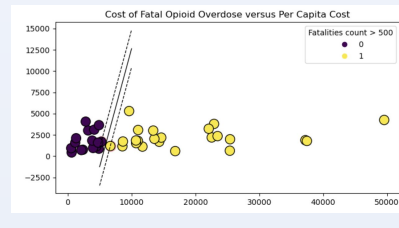
Dataset 3: Cost Component of Opioid Use Disorder and Fatal Opioid Overdose, by Jurisdiction

K-means clustering within dataset 3 revealed that there are about six groups to classify states based on costs to different programs for Opioid Use Disorder and Fatal Opioid Overdoses. There wasn't great regional clusters, but population had some correlation. Most states that are in the same cluster many wouldn't put states in the clusters they were. The Multiple Linear Regression Model demonstrated that Criminal Justice and Lost Productivity costs significantly affect the amount of cases per jurisdiction.



Dataset 4: Per Capita Cost and Overdose
SVM on dataset 4 revealed there is a linear separation between the fatal opioid overdose rate and the per capita cost. Once the opioid death hits 500 the per capita and cost of fatal overdoses separates noticeably.

Hierarchical clustering showed that there are 5 distinct groupings of opioid deaths compared to per capita and overall overdose costs. They center around each band of \$10,000's.



CONCLUSIONS

States in the Northeast and parts of the Midwest have the highest overdose rates, particularly involving fentanyl and heroin. Prescribing behaviors were found to vary by provider credentials, indicating that education and training significantly influence opioid prescribing habits. A strong correlation was identified between overdose rates and healthcare costs, with patient demographics and economic conditions playing important roles in outcomes. Specific drug classes, such as fentanyl, heroin, and methamphetamine, were found to drive regional overdose patterns. Additionally, population size and economic productivity loss were shown to be major contributors to the financial and healthcare burden associated with the opioid crisis.

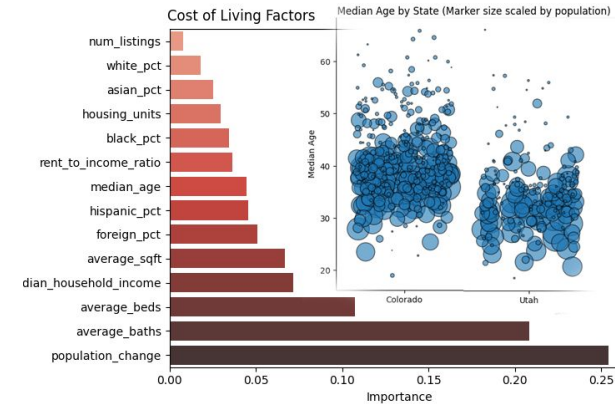
REFERENCES

- <https://data.cms.gov/summary-statistics-on-use-and-payments/medicare-medicaid-opioid-prescribing-rates/medicaid-opioid-prescribing-rates-by-geography>
- <https://www.cdc.gov/nchs/nvss/vsrr/prov-drug-involved-mortality.htm>
- <https://www.samhsa.gov/data/data-we-collect/nsduh-national-survey-drug-use-and-health/state-releases>
- <https://www.cdc.gov/mmwr/volumes/70/wr/mm7015a1.htm>

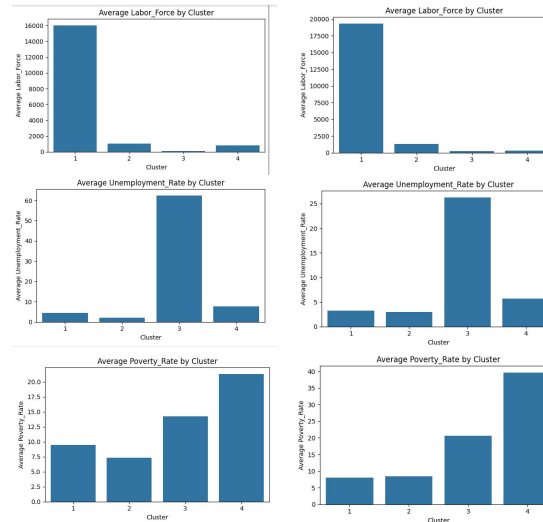
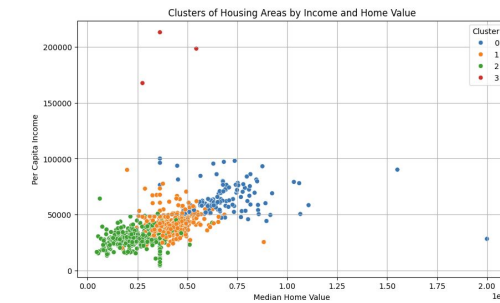
Group 2: Lifestyle Comparison between Colorado and Utah

[Website](#) [Github](#) Team Member: Lingyi Wang, Riley Jones, and Taihei Sone

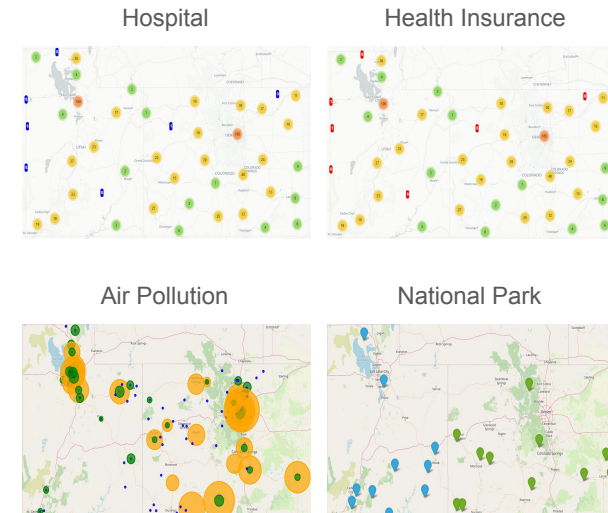
Our Mission: To provide a comprehensive, data-driven comparison of lifestyles in Colorado and Utah.



- Utah ZIPs are younger than Colorado's even after adjusting for size
- Housing price is driven by structure like beds baths and growth rather than income or race
- Migration follows housing supply not deeper socioeconomic factors



- Map Visualization in both states
 - # of hospital
 - % of insurance coverage
 - Air pollution levels
 - National parks locations
- Visually compared the medical facilities and natural environment in both states at the postcode level.





US Water Quality - PFAS Contaminants in Drinking Water

Chenchen Yuan, Arwyn Lewis, Copeland Laris, Moritz Knoedler | University of Colorado, Boulder - Dept. of Engineering, Data Science | Spring 2025

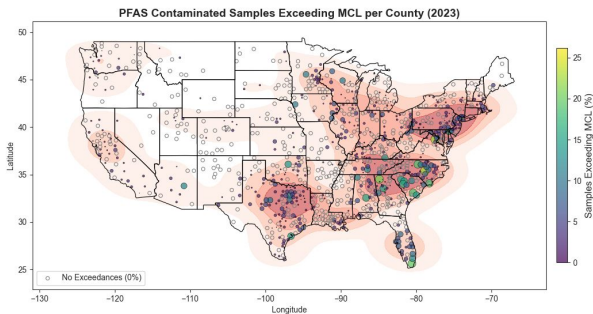


Overview + Motivation

- **PFAS = “Forever Chemicals”**: found in industrial products, linked to health risks (if > MCL)
- **Motivation**: Analyze contamination patterns, socioeconomic links, and build predictive models
 - Which **regions** are most affected?
 - Does **poverty** predict contamination?
 - Which PFAS **co-occur** most frequently?
 - Can we **predict MCL violations** using ML?

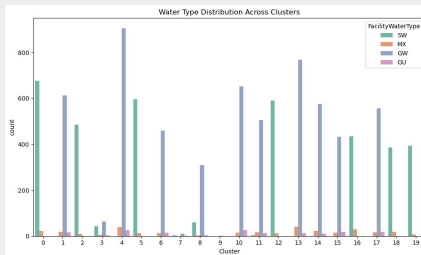
Data Preparation

- **Sources**: EPA for PFAS sample data, U.S. Census SAIGE for poverty and income (> 1 Mio water samples)
- Merged dataset matching Public Water Systems to US counties using Google Places API (focus on year 2023)

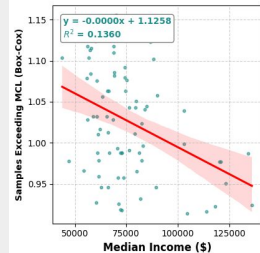
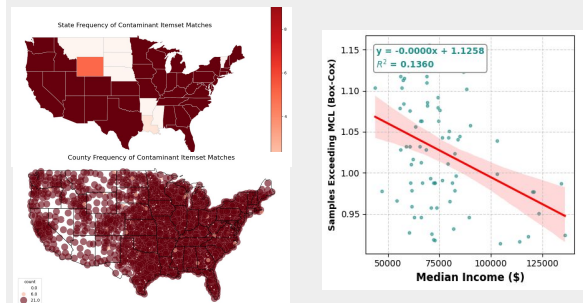


Challenges in Data Modeling

- Frequent Pattern Mining: **Apriori**
 - Long runtime, formatting data properly
 - Tuning support, confidence, and lift properly
- Classification: **SVM, Naive Bayes, Decision Trees**
 - Model specific data formatting
 - Parameter tuning for performance optimization and overfit reduction
 - Fairness audit
- Clustering: **KMeans Clustering**
 - KMeans clustering assumes spherical clusters, which are uncommon in environmental and socioeconomic data
 - Requires a predetermined number of “k” clusters
- Regression: **Linear BoxCox Regression**
 - Non-normal distribution of residuals, BoxCox transformation significantly improves model
 - Influential outliers (Cook's distance)



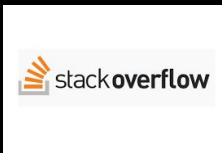
Key Findings



- KMeans: **High-contamination clusters** are associated with **groundwater sources** and **low income regions**
- Lin. Reg.: Contamination levels and ratio of **samples exceeding MCL correlate with income** ($R^2 = 0.14$)
- **County-level Apriori** modeling provided the clearest patterns of co-occurring contaminants across the U.S.
- **PFHpA, PFHxA, and PFBS** were the **most common PFAS** exceeding MCL, reflecting legacy long-chain PFAS and their short chain byproducts
- **Best classification model**: RBF SVM achieved the **highest accuracy (90%)** in predicting whether a sample exceeds MCL
- **Fairness Audit** conducted on the SVM model showed that the mode **under-performed** when identifying water samples that exceeded the MCL level **in high-poverty regions**

Analysis on Developers

Group 4, Adrian Nica, TJ Jablonski, Assaf Boneh



[GitHub](#)
[Website](#)

Overview

This project aims to find trends in job satisfaction and compensation in developers by applying machine learning and pattern mining techniques onto Overflow Developer Survey. The goal is to provide meaningful insights that can help employers, and developers better understand what drives satisfaction and compensation in the tech industry.

Challenges in Data Processing/Collection

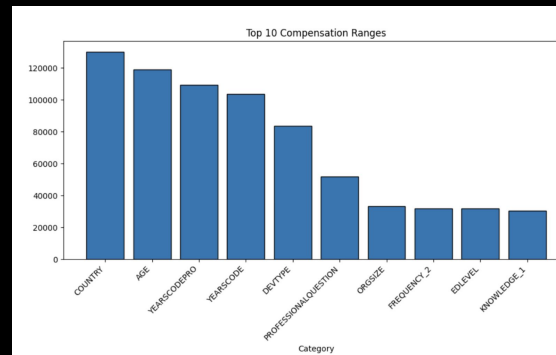
Combining multiple years of developer survey data came with challenges—mainly standardizing formats and making everything consistent. With so many columns, it was tough to manage, so we had to manually go through the survey and discuss which ones to keep or drop.

Challenges in Model Selection and Tuning

Early versions of the dataset included many irrelevant or noisy features, which led to poor performance. We also found that, due to the size of the dataset, training certain models or performing extensive hyperparameter tuning became computationally expensive. This required us to prioritize models that offered a balance between performance and efficiency.

What We Learned

This project highlighted the importance of thoughtful data preprocessing, especially when working with large and complex survey data. We saw firsthand how model performance can vary based on data distribution. Hyperparameter tuning was essential for improving accuracy, complexity costs, and other key metrics. Ultimately, balancing model complexity, runtime, and interpretability we were able to create valuable insights about the tech industry.

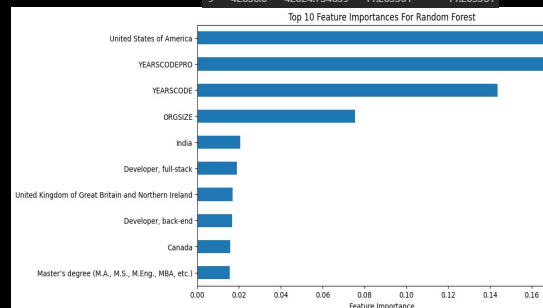


Frequency_2 ("How often do you interact with people outside of your immediate team?")

Knowledge_1 ("I have interactions with people outside of my immediate team.")

Random Forest Predictions

	True	Predicted	Error	Absolute Error
0	85672.0	85672.111927	0.111927	0.111927
1	15600.0	15601.247793	1.247793	1.247793
2	10776.0	10778.271199	2.271199	2.271199
3	70771.0	70768.624729	-2.375271	2.375271
4	162000.0	162002.558167	2.558167	2.558167
5	41448.0	41442.217494	-5.782506	5.782506
6	6934.0	6927.472588	-6.527412	6.527412
7	107937.0	107945.664080	8.664080	8.664080
8	13716.0	13706.368726	-9.631274	9.631274
9	42836.0	42824.734639	-11.265361	11.265361



Group 09: Using Machine Learning to analyze Public Transportation effect on Environmental Sustainability

Aloken Chaudhari & Joel Hoversten

Website: <https://sites.google.com/colorado.edu/5502-chaudhari-hoversten/home>

GitHub Repository: <https://github.com/joelhov09/CSCI-5502/blob/main/Milestone%202>

Overview/Motivation

Public transport and its accessibility have been greatly increasing in the recent years. This issue is something highly specific to regions and cities in the US, where certain cities have higher access and infrastructure in place for public transport than others. The initiatives for some of the key decision-makers in politics when it comes to carbon emissions could be heavily influenced by analysis like this. Insights into these factors would allow us to draw conclusions about which cities have the most sustainable models.

Data Collection/Preprocessing

Our data comes from the U.S. Energy Information Administration website and a webscraped dataset from the Bureau of Transportation Statistics. The EIA site offers a registration for an API that we used to get data on the gas consumption per state. This API provided the state and the corresponding Production, Consumption per Capita and Expenditures per Capita for each state. The original dataset was at the city level and we merged it at the state level with the EIA dataset to analyze. This dataset gave us the types of transportation offered in each state and how many of each was offered. The API was also used to get CO2 emissions for the state.

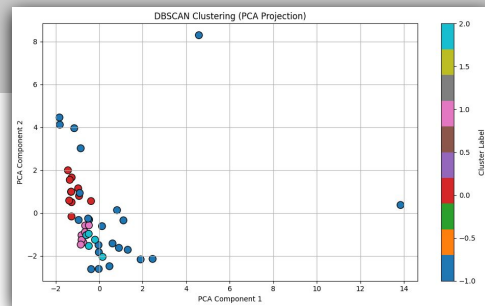
Data Modeling

FP-Growth- states with above-median public transportation access were frequently associated with below-median energy consumption per capita

SVM- Linear Kernel misclassified 3 out of 14 and properly classified 11 out of 14 in our dataset.

DBSCAN - Works good for our Data since their is high variance. Came out to 4 clusters

Logistic Regression- performed worse than SVM misclassified 5 out of 14 and properly classified 9 out of 14 in our dataset.



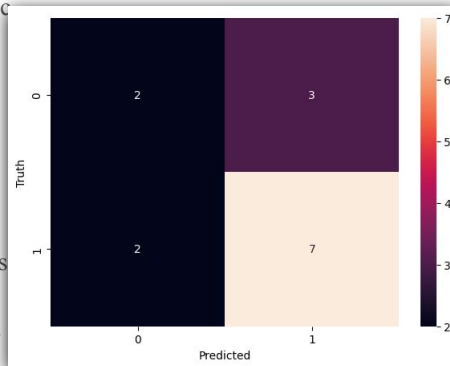
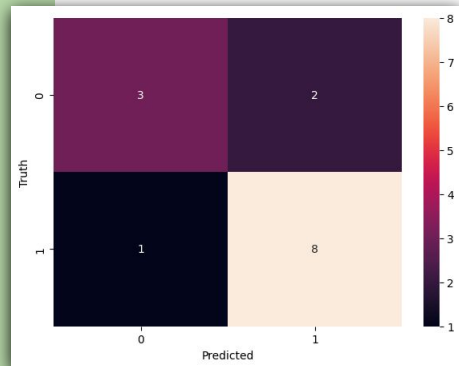
Future Work

Model Enhancements-

Current models reliance on relatively simple models, such as SVMs and Logistic Regression. Future studies could explore ensemble learning techniques such as Random Forests or XGBoost. For identifying intricate feature interactions and improved classification performance.

Dataset Additions-

The analysis could be enhanced with additional datasets that capture political, urban, and socioeconomic dimensions more precisely. granular urban data, emissions by transportation mode



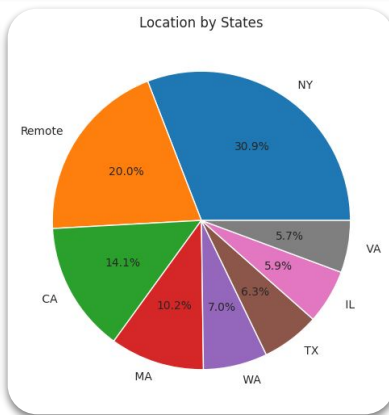
GROUP 10 : Job Search Exploration

[Webpage](#) | [GitHub](#)

Hiromasa Ito ◦ Chien Lin ◦ Mai Fukushima

Objective and Motivation

This project analyzes employment trends in data science and analytics by examining job postings, focusing on salary structures, skill requirements, and the impact of demographic factors such as race, ethnicity, and gender. The goal is to help job seekers, employers, and policymakers make informed decisions and promote a more transparent and equitable labor market.



Key Insights

- Our models can be applied for predictions for job listings, fake job postings, job titles, building job recommendation platforms.
- Found relationships between Salary and Years of Experience
- Analyzing the key features, provide users with highly accurate and relevant job suggestions

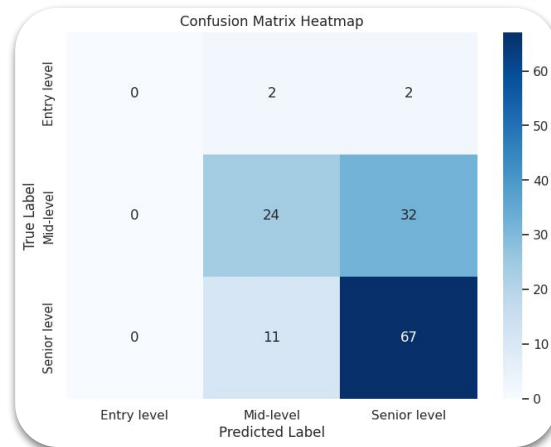
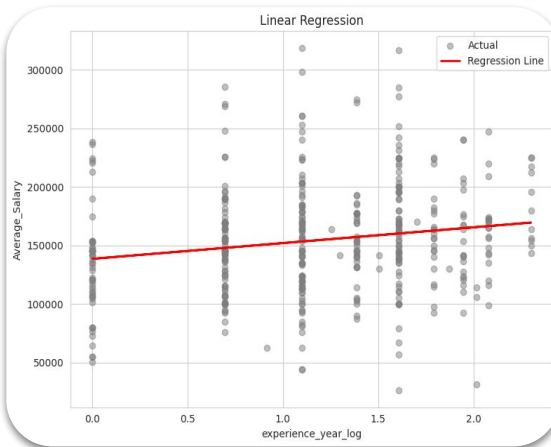
Challenges

Data Preprocessing:

- Web scraping data from the internet
- Skewness, outliers, removing duplicates, scaling, transforming

Data Modeling:

- Choosing the right model
- Setting up data for model implementation
- Tuning the model
- Reviewing the results (accurate, flaws, bias, etc.)



Group 11: Exploring Education on a Global Scale - OECD PISA 2022 Insights

Sean Fitzgerald, Nick Mobus and Vanessa Thorsten

Website URL: <https://wix.to/Gfvli7L> | GitHub URL: <https://github.com/vt-art/OECD-PISA-2022-Insights>

INTRODUCTION

Our research aims to draw insights from the 2022 Programme for International Student Assessment (PISA), an academic test and survey of 15-year-old students by the Organisation for Economic Co-operation and Development (OECD).

METHODS

In 2022, over half a million students from 81 countries completed the PISA. We rolled-up student-level responses to combine these data with country-level PISA scores and GDP / % GDP spent on education.

PISA 2022
student questionnaires
(N = 613,744 students)



PISA 2022 math
scores & socio-
economic status
(N = 81 countries)

Roll-Up-
weighted
averages
(N = 81 countries)

Gross Domestic
Product (GDP) -
World Bank (WB) 2021
(N = 80 countries)

% GDP spent on
education -
WB 2019-2023
(N = 75 countries)

We cleaned the data, created new + scaled features & addressed missing values. After investigating 10 topics through EDA, we focused on the following:

- continuous math score & literacy (linear regression),
- binary minimum math proficiency & predictive features (naïve Bayesian & support vector machines (SVM)), and
- math proficiency level with food insecurity through random forest (RF) & K-Nearest Neighbors (KNN)

For each model, we chose the best features, tuned hyper-parameters, & evaluated metrics of model fit + performance. For each question, we determined the best model and interpreted the findings.

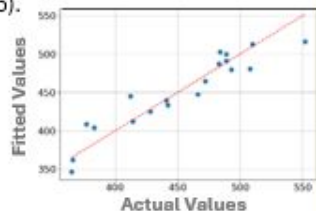
RESULTS

Q1: Math Proficiency & Literacy:

During EDA, math proficiency plotted against reading proficiency showed a strong, linear, positive relationship. Utilized linear regression to

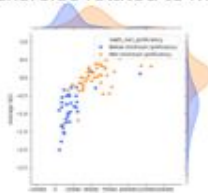
investigate the relationship further.

Additionally, we included other behavioral variables as well. Adj. R^2 and RMSE were used to select the best model, which included mean reading score and a love of learning score as parameters. (Adj. $R^2 = 0.894$, RMSE = 18.95).



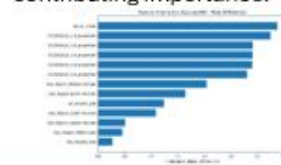
Q2: Predictors of Minimum Math Proficiency:

Binned math score [$<$ min standards (<420), met minimum standards ($420 <$), EDA identified education spending, mean SES, and questions related to food security and exercise related to math proficiency.



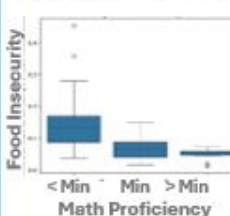
GNB classified countries with $<$ min average scores (precision = 0.71, recall = 0.83

& F1 = 0.76). AUC = 0.96. Mean SES and food security identified as most important, with some geographic regions also contributing importance.



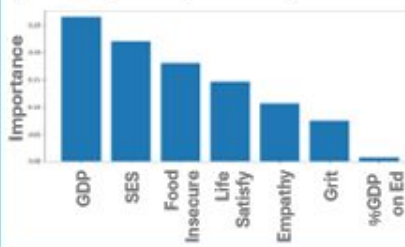
Q3: Math Scores & Food Security:

Binned math score [$<$ min standards (<420), met min ($420 <$ <482), & $>$ min ($482+$)]. EDA found higher food insecurity related to poorer scores.



RF classified countries with $<$ min average scores well (precision=0.8 recall=0.89, & f1-score=0.84)

AUC one-vs-rest = 0.87. GDP, SES & food insecurity identified as important features in predicting math proficiency level.



CONCLUSIONS

Proficiency scores are highly predictive of each other, however no other behavioral variables had significant predictive power. Higher math proficiency corresponds with greater economic measures, such as the country's GDP, average student SES and average student food security. These findings highlight the need for economic stability & funding in education success.